

Predicting and correcting springback in robotic incremental sheet forming

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[ALL RESULTS: SIMULATION](#)

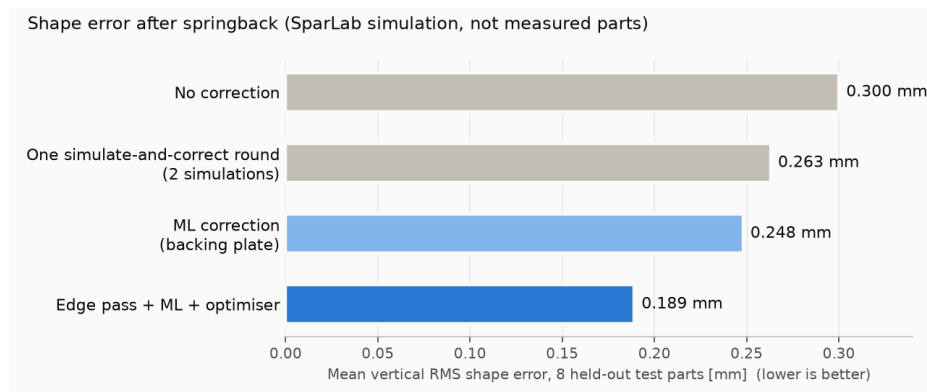
PROBLEM

In incremental sheet forming a robot pushes a clamped sheet into shape one contour at a time. When the tool and clamps let go, the elastic part of the deformation recovers and the part **springs back**, coming out shallower and bent. Today each new geometry is fixed by iteration: form, scan, correct the toolpath, form again. Every loop costs a sheet, robot time and a scan. The goal of this project is to move as much of that loop as possible into software, so the first formed part lands close to the target.

APPROACH

- **Physics: a forming simulator written from scratch in C++.** Implicit finite-element model with moving rigid tools, frictional contact, anisotropic (Hill48) and kinematic-hardening (Chaboche) plasticity at finite strain, thin-sheet elements with 5 integration points through the thickness, and the full sequence of forming, unloading and release onto 3-point supports, where springback happens.
- **Learning: an ML surrogate trained on the simulations.** A neural-network ensemble predicts the springback at every point of a part in seconds, with an uncertainty estimate and a check that flags parts unlike its training data.
- **Correction: an optimiser on the surrogate** searches the commanded shape whose predicted result best matches the target, then one simulation verifies it.
- **Process change where software cannot reach.** Most of the remaining error sat at the top edge next to the clamp, which toolpath changes cannot move. A second tool sweeping that edge from underneath removed it, and the model learned the new process from only 8 extra simulations through a transfer correction rather than a full retrain.

RESULTS (8 HELD-OUT TEST PARTS, SIMULATION)



Mean vertical RMS deviation from the target after release. Domes, cones, elliptic cones and pyramids, 15 to 20 mm across, 1 mm AA5754-O sheet; none seen in training.

- **37% less shape error than an uncorrected part** (0.300 to 0.189 mm), on the first attempt, and better than a full simulate-and-correct round that needs twice the simulations.
- Better than the plain ML correction on 7 of 8 parts; the best part improved from 0.307 to 0.071 mm. The edge error fell from 0.45 to 0.13 mm.
- The surrogate predicts springback to within 0.047 mm on unseen parts, against a typical springback of 0.29 mm, and runs in seconds instead of the 45 to 90 minutes of a simulation.

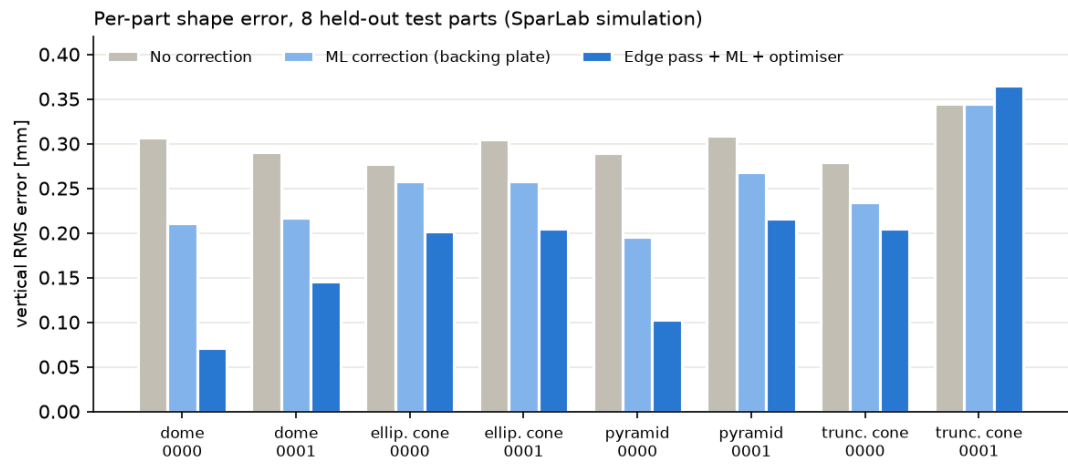
LIMITS

All results are simulations; nothing has yet been compared with a formed and scanned part. Parts are small (15 to 20 mm) with one material and one tool. Eight test parts is a small sample, one part got slightly worse, and the optimiser is optimistic about its own result (it predicts 0.14 mm against 0.19 mm simulated).

NEXT STEPS

- **Validate on real data:** compare one formed and scanned part with the simulation, then fit the transfer correction to scans, which the pipeline already supports.
- **Close the loop automatically:** ML first shot, check, and one engine correction only when a part misses tolerance (implemented; being evaluated).
- **Scale:** larger parts, more materials, and faster simulations to grow the training set.

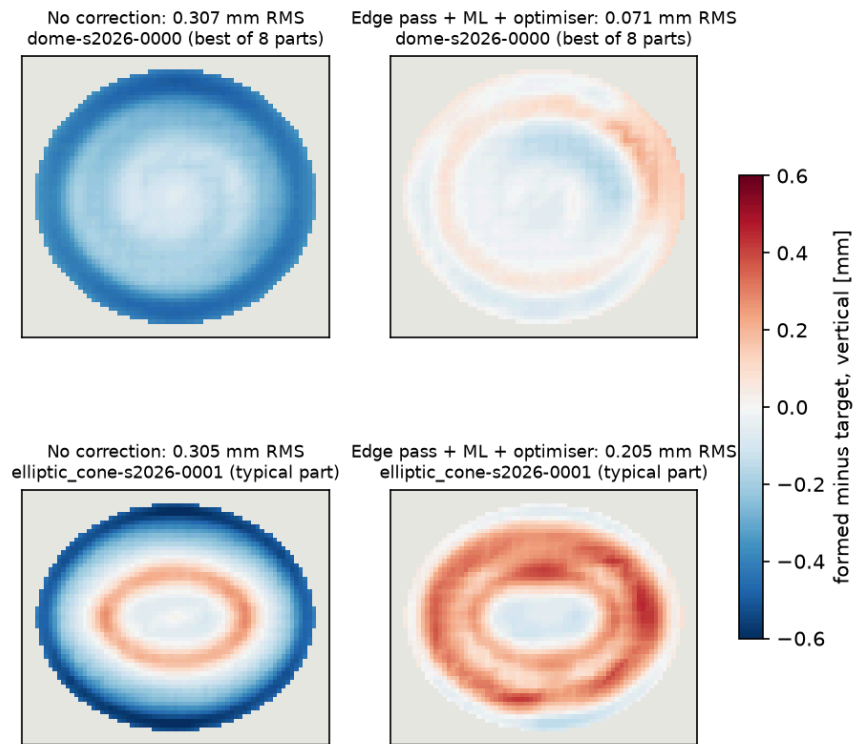
APPENDIX A. PER-PART RESULTS



Vertical RMS deviation over the part for each held-out test part. One verifying simulation per part and method.

Method	Mean error	Simulations per part	Better than no correction
No correction	0.300 mm	1	-
One simulate-and-correct round	0.263 mm	2	7 of 8
ML correction (backing plate)	0.248 mm	1	8 of 8
Edge pass + ML + optimiser	0.189 mm	1	7 of 8

APPENDIX B. DEVIATION MAPS



Formed minus target, vertical, over the part (the flat flange outside the part is not part of the error metric and is shown in grey). Blue: too deep; red: too shallow. Top row: the best of the 8 parts; bottom row: a typical part.

APPENDIX C. THE ENGINE

Forming simulator (C++17)

- Implicit quasi-static finite elements; Newton solver with sparse direct factorisation (SuiteSparse CHOLMOD / UMFPACK).
- Rigid sphere, plane and cylinder tools on tabulated trajectories; penalty contact with Coulomb friction.
- Plasticity: von Mises, Hill48 anisotropy, Chaboche kinematic hardening, finite strain through a logarithmic-strain formulation.
- Incompatible-mode hexahedra (no bending locking in thin sheet) with 5 integration points through the thickness.
- Steps: form, unload, optional edge pass by the underside tool, release onto 3-2-1 supports; restartable state between steps.
- Built on a structural FEA and topology-optimisation engine with a verification suite (patch tests, analytical benchmarks, cross-checks against independent codes). The forming mesh was chosen from a mesh-convergence study.

Pre-compensation and ML (Python)

- Part families, toolpath generation (ball-tool offset, contour and spiral paths), simulation decks and a content-addressed run cache.
- 57 geometric, process and material features per surface point.
- Neural-network ensemble (5 networks) and gradient-boosted ensemble (8 models) with per-point uncertainty; conformal prediction intervals.
- Out-of-distribution check that refuses parts outside the training envelope.
- Transfer correction to adapt to a new process or to scans from a few samples.
- Surrogate optimiser with physical limits: wall angle, tool reach, held regions.
- Trained on 72 simulated parts (144 forming simulations), calibrated on 24, tested on 8 never seen in training.

APPENDIX D. EVALUATION PROTOCOL AND WHAT DID NOT WORK

- Test parts are fixed in advance and never used for training, model selection or tuning. Every reported error comes from a full simulation of the corrected toolpath.
- More training data improved the surrogate's prediction error (0.052 to 0.047 mm) but not the corrected parts, which showed the limit was the correction strategy, not the data. That finding led to the edge pass.
- A follow-up that retrained on the optimiser's own shapes and penalised uncertainty more strongly made the predictions more realistic (gap 0.05 to 0.02 mm) but the parts slightly worse (0.203 mm). It is recorded and being separated into its two causes.
- Code, data provenance, and every benchmark table are in the repository, with a reproduction script for each result.